

Improving Preparedness of Communities for Evacuations using ZEVs

(Task 2: Optimizing Power System Resilience)

Saeed Aliamooei Lakeh

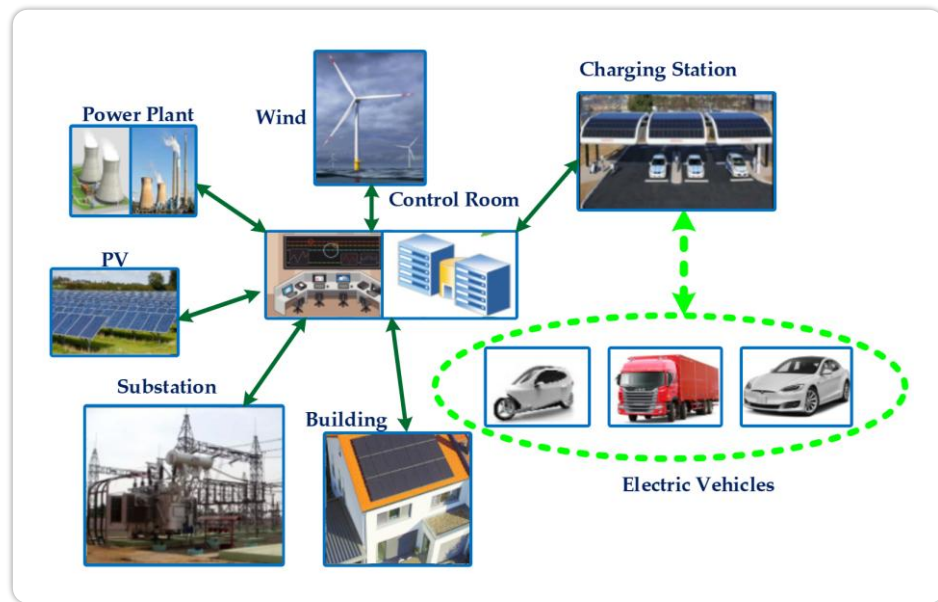
Fall 2025: October 02, 2025

Workshop #3 at UC Merced

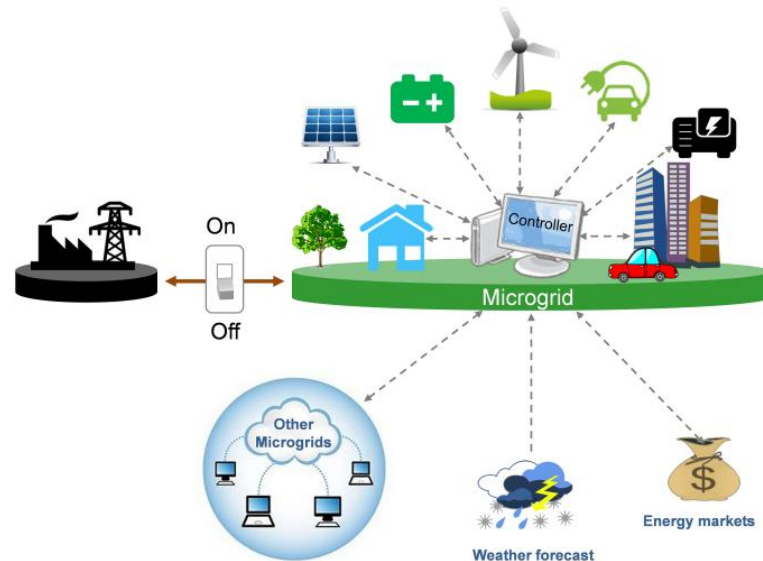
Contents

- Introduction
- Power Systems Optimization and EVs: Optimal Power Flow
- Mariposa Power Grid: GIS Synthetic grid
- Transportation and Evacuation Layer
- Conclusion and Future Plan

Power Systems



Power System



Copyright Berkeley Lab

Microgrid

IEEE RTS-79

24-bus system 11 generators

IEEE Transactions on Power Apparatus and Systems,
Vol. PAS-98, No.6 Nov./Dec. 1979. IEEE
RELIABILITY TEST SYSTEM

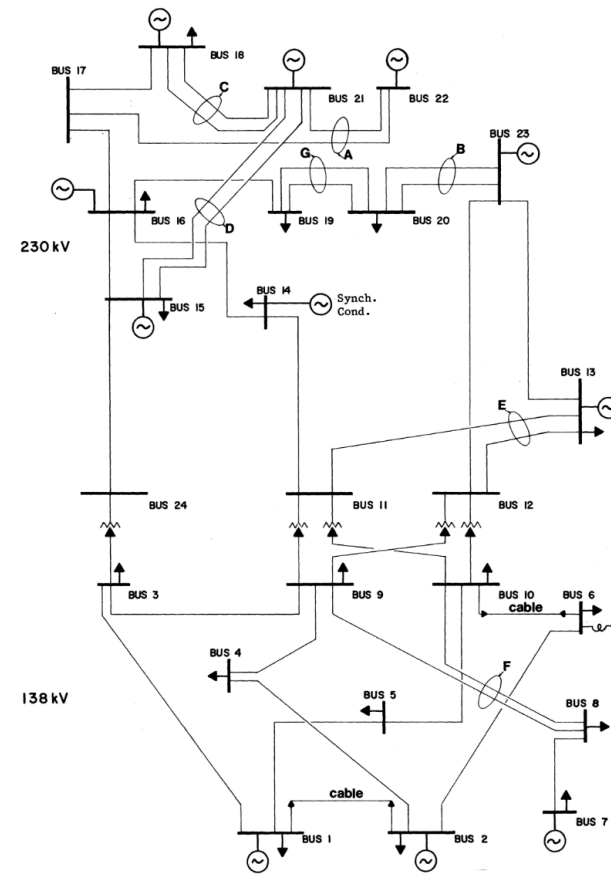


Figure 1 - IEEE Reliability Test System

MATPOWER (MATLAB)

Command Window

```
>> case24_ieee_rts
```

```
ans =
```

```
struct with fields:
```

```
version: '2'
baseMVA: 100
bus: [24x13 double]
gen: [33x21 double]
branch: [38x13 double]
gencost: [33x7 double]
```

```
>> runopf(case24_ieee_rts)
```

```
MATPOWER Version 8.0, 17-May-2024
```

```
Optimal Power Flow -- AC-polar-power formulation
```

```
MATPOWER Interior Point Solver -- MIPS, Version 1.5.1, 10-May-2024
```

```
(using built-in linear solver)
```

```
Converged!
```

```
OPF successful
```

```
Converged in 2.58 seconds
```

```
Objective Function Value = 63352.21 $/hr
```

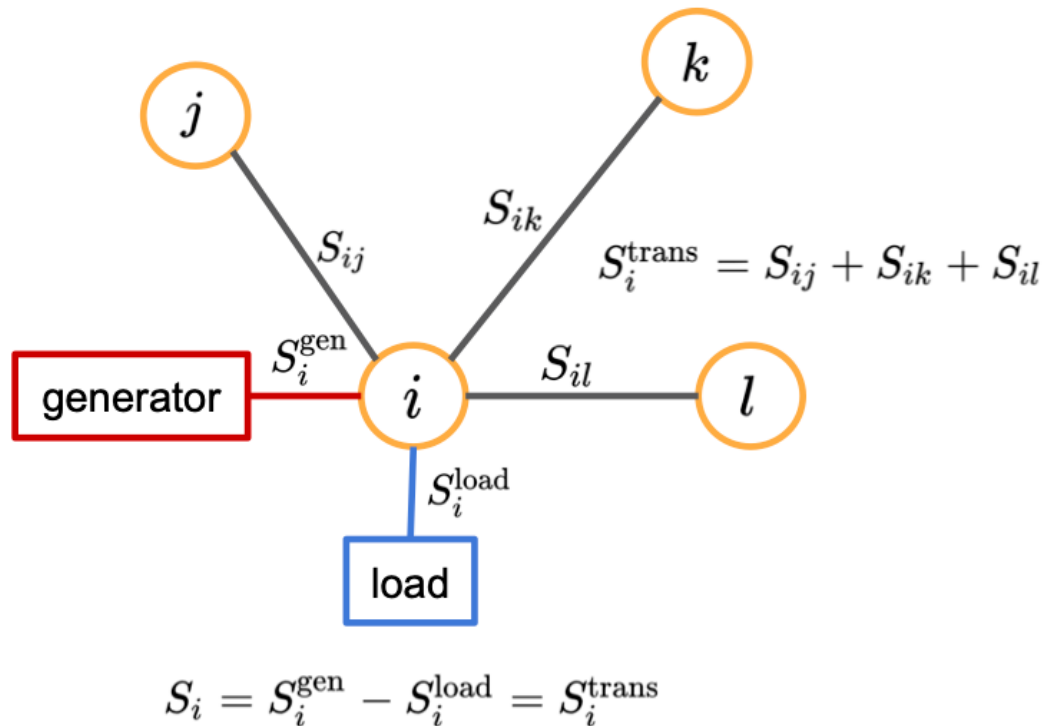
System Summary

How many?	How much?	P (MW)	Q (MVar)
Buses	24	Total Gen Capacity	3405.0
Generators	33	On-line Capacity	3405.0
Committed Gens	33	Generation (actual)	2896.8
Loads	17	Load	2850.0
Fixed	17	Fixed	2850.0
Dispatchable	0	Dispatchable	-0.0 of -0.0
Shunts	1	Shunt (inj)	-0.0
Branches	38	Losses ($I^2 * Z$)	46.77
Transformers	5	Branch Charging (inj)	-
Inter-ties	10	Total Inter-tie Flow	1319.0
Areas	4		

	Minimum	Maximum
Voltage Magnitude	1.006 p.u. @ bus 24	1.050 p.u. @ bus 10
Voltage Angle	-12.55 deg @ bus 8	20.69 deg @ bus 22
P Losses ($I^2 * R$)	-	6.09 MW @ line 14-16
Q Losses ($I^2 * X$)	-	47.41 MVar @ line 14-16
Lambda P	45.24 \$/MWh @ bus 22	52.43 \$/MWh @ bus 8
Lambda Q	-0.44 \$/MWh @ bus 10	0.75 \$/MWh @ bus 3

Optimal Power Flow

- Constrained optimization.
- Find the best way to operate the grid by determining control variables to satisfy a specific objective.
- Objectives: Minimizing costs, minimizing emission, minimizing power loss, subject to physical and engineering constraints



Optimal Power Flow

Bus Data									
Bus #	Voltage		Generation		Load		Lambda (\$/MVA-hr)		
	Mag (pu)	Ang (deg)	P (MW)	Q (MVar)	P (MW)	Q (MVar)	P	Q	
1	1.047	-6.755	184.00	6.56	108.00	22.00	49.588	-	
2	1.047	-6.831	184.00	3.77	97.00	20.00	49.612	-	
3	1.014	-6.187	-	-	180.00	37.00	49.687	0.749	
4	1.016	-9.574	-	-	74.00	15.00	51.123	0.363	
5	1.035	-9.737	-	-	71.00	14.00	50.851	-0.065	
6	1.032	-12.310	-	-	136.00	28.00	51.819	-0.254	
7	1.036	-9.877	211.03	49.32	125.00	25.00	51.072	-	
8	1.009	-12.553	-	-	171.00	35.00	52.425	0.329	
9	1.024	-7.818	-	-	175.00	36.00	50.398	0.263	
10	1.050	-9.678	-	-	195.00	40.00	50.657	-0.444	
11	1.023	-2.613	-	-	-	-	50.273	0.035	
12	1.017	-1.876	-	-	-	-	50.173	0.090	
13	1.033	0.000*	235.74	97.80	265.00	54.00	49.707	-	
14	1.040	1.012	0.00	114.92	194.00	39.00	49.454	-	
15	1.041	9.324	167.00	109.99	317.00	64.00	47.643	0.140	
16	1.044	8.543	155.00	80.00	100.00	20.00	47.805	0.094	
17	1.048	12.892	-	-	-	-	46.865	0.061	
18	1.050	14.247	400.00	72.89	333.00	68.00	46.575	-	
19	1.039	7.493	-	-	181.00	37.00	48.045	0.107	
20	1.044	8.452	-	-	128.00	26.00	47.834	0.052	
21	1.050	15.023	400.00	-7.45	-	-	46.411	-	
22	1.050	20.691	300.00	-38.48	-	-	45.239	-	
23	1.050	9.684	660.00	45.63	-	-	47.564	-	
24	1.006	3.668	-	-	-	-	48.998	0.590	
Total:			2896.77	534.96	2850.00	580.00			

Branch Data									
Brnch #	From Bus	To Bus	From Bus P (MW)	Injection Q (MVar)	To Bus P (MW)	Injection Q (MVar)	Loss (I^2 * Z)		
							P (MW)	Q (MVar)	
1	1	2	10.09	-27.00	-10.09	-23.55	0.002	0.01	
2	1	3	-0.65	13.64	0.79	-19.17	0.140	0.54	
3	1	5	66.55	-2.07	-65.67	3.00	0.881	3.41	
4	2	4	44.12	13.24	-43.47	-14.38	0.651	2.51	
5	2	6	52.98	-5.92	-51.70	5.23	1.276	4.93	
6	3	9	21.25	-15.45	-21.06	12.85	0.192	0.74	
7	3	24	-202.04	-2.38	203.01	37.74	0.969	35.36	
8	4	9	-30.53	-0.62	30.78	-1.37	0.242	0.94	
9	5	10	-5.33	-17.00	5.39	14.63	0.059	0.23	
10	6	10	-84.30	-139.82	85.24	-122.71	0.937	4.08	
11	7	8	86.03	24.32	-84.84	-21.46	1.191	4.60	
12	8	9	-50.11	3.68	51.18	-4.17	1.068	4.13	
13	8	10	-36.05	-17.22	36.69	14.95	0.639	2.47	
14	9	11	-110.63	-25.77	110.93	36.73	0.300	10.96	
15	9	12	-125.26	-17.54	125.64	31.13	0.372	13.58	
16	10	11	-153.72	22.06	154.25	-2.97	0.523	19.10	
17	10	12	-168.59	31.07	169.23	-7.80	0.638	23.27	
18	11	13	-102.24	-13.13	102.85	7.36	0.613	4.79	
19	11	14	-162.94	-20.63	164.33	22.00	1.384	10.71	
20	12	13	-75.36	-28.97	75.72	21.34	0.368	2.87	
21	12	23	-219.51	5.64	225.31	17.89	5.805	45.22	
22	13	23	-207.84	15.11	212.39	0.65	4.553	35.48	
23	14	16	-358.33	53.92	364.42	-15.39	6.094	47.41	
24	15	16	82.03	-31.18	-81.88	28.43	0.154	1.21	
25	15	21	-218.92	14.60	221.73	-4.00	2.809	21.85	
26	15	21	-218.92	14.60	221.73	-4.00	2.809	21.85	
27	15	24	205.81	47.98	-203.01	-37.74	2.798	21.67	
28	16	17	-315.32	36.12	318.37	-18.10	3.055	23.98	
29	16	19	87.77	10.84	-87.55	-14.43	0.217	1.67	
30	17	18	-179.82	5.23	180.35	-4.32	0.531	4.25	
31	17	22	-138.55	12.87	140.99	-18.18	2.438	19.02	
32	18	21	-56.68	4.61	56.77	-9.85	0.098	0.77	
33	18	21	-56.68	4.61	56.77	-9.85	0.098	0.77	
34	19	20	-46.72	-11.28	46.83	3.07	0.105	0.82	
35	19	20	-46.72	-11.28	46.83	3.07	0.105	0.82	
36	20	23	-110.83	-16.07	111.15	13.55	0.320	2.47	
37	20	23	-110.83	-16.07	111.15	13.55	0.320	2.47	
38	21	22	-157.00	20.25	159.01	-20.30	2.007	15.64	
Total:							46.766	416.60	

MATPOWER

Voltage Constraints

Bus #	Vmin mu	Vmin	V	Vmax	Vmax mu
1	-	0.950	1.047	1.050	0.418
2	-	0.950	1.047	1.050	0.409
10	-	0.950	1.050	1.050	1496.967
18	-	0.950	1.050	1.050	365.299
21	-	0.950	1.050	1.050	1108.677
22	-	0.950	1.050	1.050	662.334
23	-	0.950	1.050	1.050	797.284

Command Window

Generation Constraints

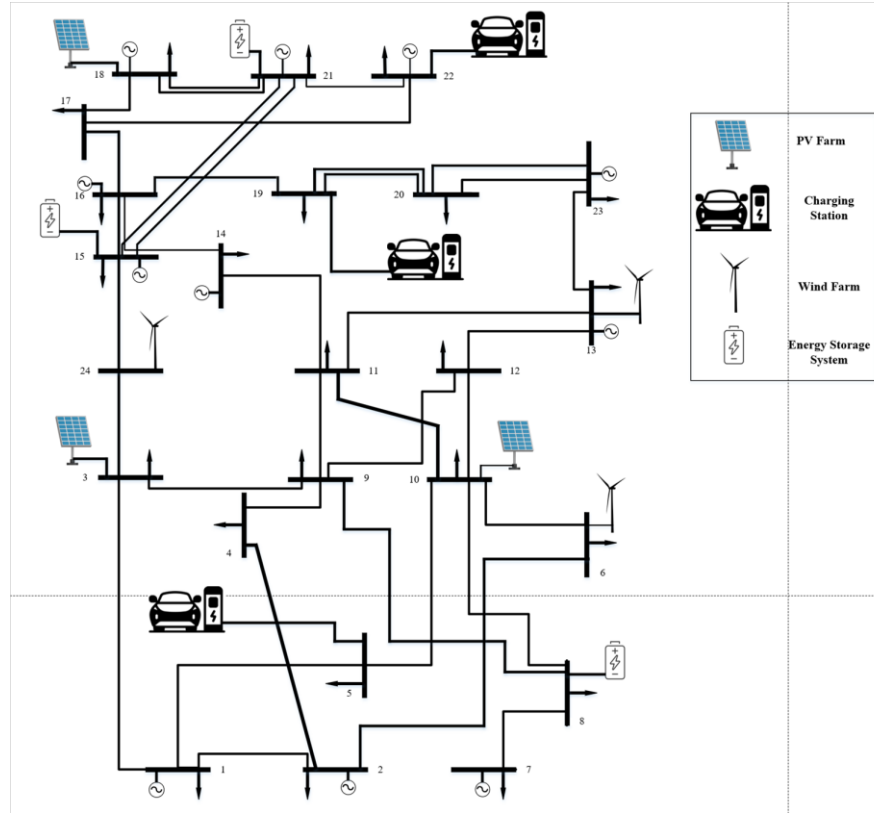
Gen #	Bus #	Active Power Limits				
		Pmin mu	Pmin	Pg	Fmax	Pmax mu
1	1	80.412	16.00	16.00	20.00	-
2	1	80.412	16.00	16.00	20.00	-
3	1	-	15.20	76.00	76.00	31.357
4	1	-	15.20	76.00	76.00	31.357
5	2	80.388	16.00	16.00	20.00	-
6	2	80.388	16.00	16.00	20.00	-
7	2	-	15.20	76.00	76.00	31.382
8	2	-	15.20	76.00	76.00	31.382
15	14	0.000	0.00	-	0.00	49.454
16	15	10.497	2.40	2.40	12.00	-
17	15	10.497	2.40	2.40	12.00	-
18	15	10.497	2.40	2.40	12.00	-
19	15	10.497	2.40	2.40	12.00	-
20	15	10.497	2.40	2.40	12.00	-
21	15	-	54.30	155.00	155.00	32.669
22	16	-	54.30	155.00	155.00	32.831
23	18	-	100.00	400.00	400.00	41.982
24	21	-	100.00	400.00	400.00	41.817
25	22	-	10.00	50.00	50.00	45.238
26	22	-	10.00	50.00	50.00	45.238
27	22	-	10.00	50.00	50.00	45.238
28	22	-	10.00	50.00	50.00	45.238
29	22	-	10.00	50.00	50.00	45.238
30	22	-	10.00	50.00	50.00	45.238
31	23	-	54.30	155.00	155.00	32.589
32	23	-	54.30	155.00	155.00	32.589
33	23	-	140.00	350.00	350.00	32.288

Gen #	Bus #	Reactive Power Limits				
		Qmin mu	Qmin	Qg	Qmax	Qmax mu
16	15	-	0.00	6.00	6.00	0.141
17	15	-	0.00	6.00	6.00	0.141
18	15	-	0.00	6.00	6.00	0.141
19	15	-	0.00	6.00	6.00	0.141
20	15	-	0.00	6.00	6.00	0.141
21	15	-	-50.00	80.00	80.00	0.140
22	16	-	-50.00	80.00	80.00	0.094

Modified 24-bus Power System

Single Line
Diagram

IEEE
Reliability
Test System
(RTS)



Python: Pyomo Code

```
# Step 2: Create 12 buses at 110 kV
bus0 = pp.create_bus(net, vn_kv=110, name="Bus 0")
bus1 = pp.create_bus(net, vn_kv=110, name="Bus 1")
bus2 = pp.create_bus(net, vn_kv=110, name="Bus 2")
bus3 = pp.create_bus(net, vn_kv=110, name="Bus 3")
bus4 = pp.create_bus(net, vn_kv=110, name="Bus 4")
bus5 = pp.create_bus(net, vn_kv=110, name="Bus 5")
bus6 = pp.create_bus(net, vn_kv=110, name="Bus 6")
bus7 = pp.create_bus(net, vn_kv=110, name="Bus 7")
bus8 = pp.create_bus(net, vn_kv=110, name="Bus 8")
bus9 = pp.create_bus(net, vn_kv=110, name="Bus 9")
bus10 = pp.create_bus(net, vn_kv=110, name="Bus 10")
bus11 = pp.create_bus(net, vn_kv=110, name="Bus 11")

# Step 3: Add external grid (slack generator) at Bus 0
pp.create_ext_grid(net, bus=bus0, vm_pu=1.02, name="Slack Generator")

# Step 4: Add generator at Bus 6 with Limits
gen2 = pp.create_gen(net, bus=bus6, p_mw=30, vm_pu=1.01,
min_p_mw=10, max_p_mw=60, name="Generator at Bus 6")

# Step 5: Add Linear cost function (25 €/MWh) for generator
pp.create_poly_cost(net, element=gen2, et="gen", cp1_eur_per_mwh=25)

# Step 6: Add Loads at various buses
pp.create_load(net, bus=bus2, p_mw=6.0, q_mvar=2.0, name="Load 2")
pp.create_load(net, bus=bus3, p_mw=4.0, q_mvar=1.5, name="Load 3")
pp.create_load(net, bus=bus5, p_mw=5.5, q_mvar=1.8, name="Load 5")
pp.create_load(net, bus=bus8, p_mw=7.0, q_mvar=2.2, name="Load 8")
pp.create_load(net, bus=bus10, p_mw=3.5, q_mvar=1.0, name="Load 10")
pp.create_load(net, bus=bus11, p_mw=4.2, q_mvar=1.3, name="Load 11")

# Step 7: Add 11 Lines with impedance and Limits
lines = [
    (bus0, bus1), (bus1, bus2), (bus2, bus3), (bus3, bus4), (bus4, bus5),
    (bus5, bus6), (bus6, bus7), (bus7, bus8), (bus8, bus9), (bus9, bus10), (bus10, bus11)
]
```

```
143 def create_objective_function(self):
172     print(" 2. Minimize Power Losses (AC formulation)")
173     print(" 3. Maximize Grid Resilience (voltage stability)")
174
175 def create_full_acopf_constraints(self):
176     """Create full ACOPT constraints"""
177
178     # 1. Reference bus angle constraint
179     def ref_bus_angle_rule(model, bus):
180         if bus in self.model.REF_BUSES:
181             return self.model.v_ang[bus] == 0
182         return pyo.Constraint.Skip
183     self.model.ref_bus_angle = pyo.Constraint(self.model.BUSES, rule=ref_bus_angle_rule)
184
185     # 2. Voltage magnitude Limits
186     def voltage_limits_rule(model, bus):
187         return (self.model.v_min[bus], self.model.v_mag[bus], self.model.v_max[bus])
188     self.model.voltage_limits = pyo.Constraint(self.model.BUSES, rule=voltage_limits_rule)
189
190     # 3. Generator power Limits
191     def gen_power_limits_rule(model, gen):
192         return (self.model.p_gen_min[gen], self.model.p_gen_max[gen])
193     self.model.gen_power_limits = pyo.Constraint(self.model.GENERATORS, rule=gen_power_limits_rule)
194
195     def gen_reactive_limits_rule(model, gen):
196         return (self.model.q_gen_min[gen], self.model.q_gen_max[gen])
197     self.model.gen_reactive_limits = pyo.Constraint(self.model.GENERATORS, rule=gen_reactive_limits_rule)
198
199     # 4. AC Power Flow Equations (simplified but full AC)
200     def ac_power_flow_rule(model, bus):
201         # Active power balance at each bus
202         gen_at_bus = sum(self.model.p_gen[gen] for gen in self.model.GENERATORS) # Simplified
203         load_at_bus = self.model.p_load[bus] + self.model.p_emergency_load[bus]
204         ev_at_bus = sum(self.model.p_ev_charging[ev] for ev in self.model.EV_STATIONS) # Simplified
205         storage_net = sum(self.model.p_storage_discharge[st] - self.model.p_storage_charge[st]
206                             for st in self.model.STORAGE)
207
208         # AC power flow
209         return gen_at_bus - load_at_bus - ev_at_bus + storage_net == 0
210     self.model.ac_power_flow = pyo.Constraint(self.model.BUSES, rule=ac_power_flow_rule)
211
212     # 5. Reactive Power Flow Equations
213     def reactive_power_flow_rule(model, bus):
214         # Reactive power balance at each bus
215         gen_at_bus = sum(self.model.q_gen[gen] for gen in self.model.GENERATORS) # Simplified
216         load_at_bus = self.model.q_load[bus]
217         return gen_at_bus - load_at_bus == 0
218     self.model.reactive_power_flow = pyo.Constraint(self.model.BUSES, rule=reactive_power_flow_rule)
219
```

California Test System

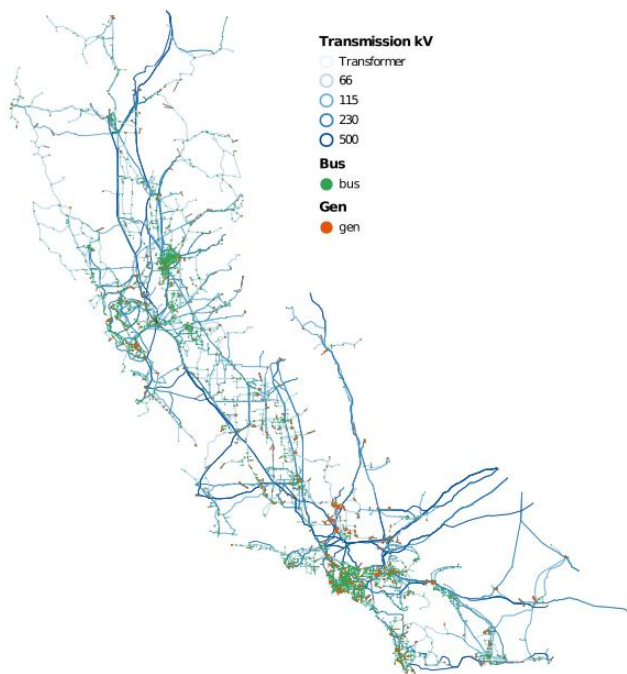


Fig. 4. California test system topology.

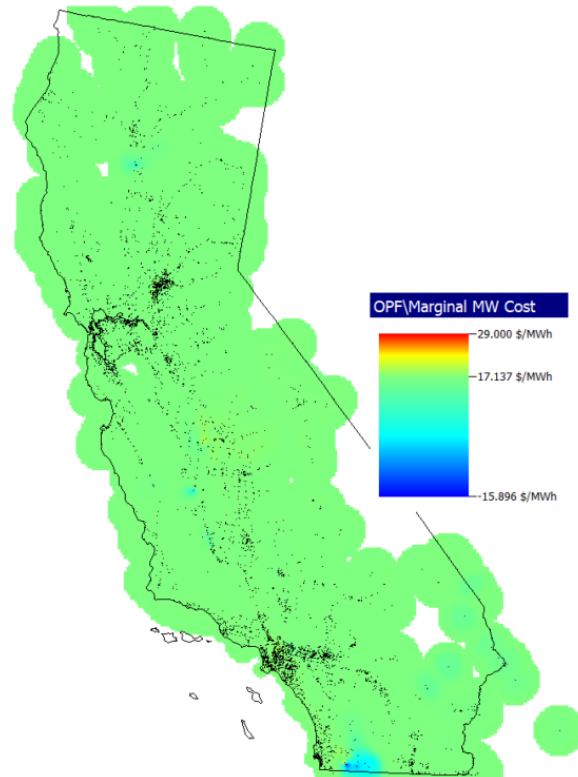


Fig. 8. LMP contour map for peak load scenario.

Taylor, Sofia, et al. "California test system (CATS): A geographically accurate test system based on the California grid." *IEEE Transactions on Energy Markets, Policy and Regulation* 2.1 (2023): 107-118.

Synthetic Grid

- Use a standard system and customize it
- Synthetic but Realistic Distribution

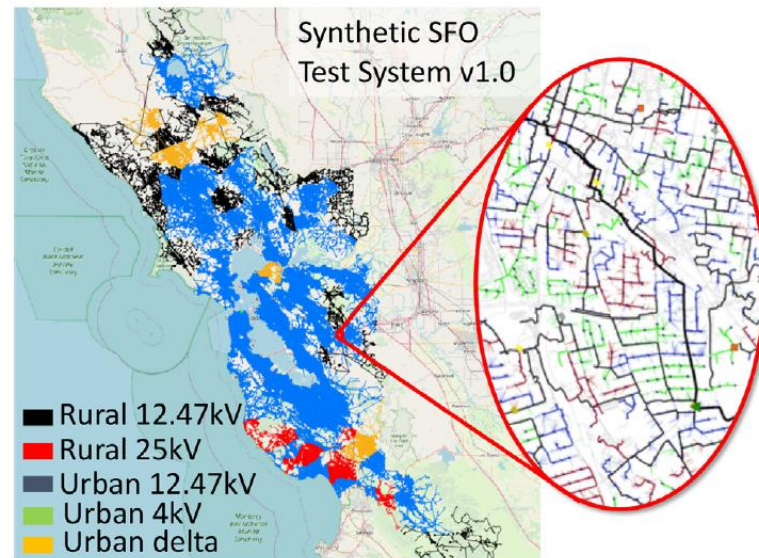
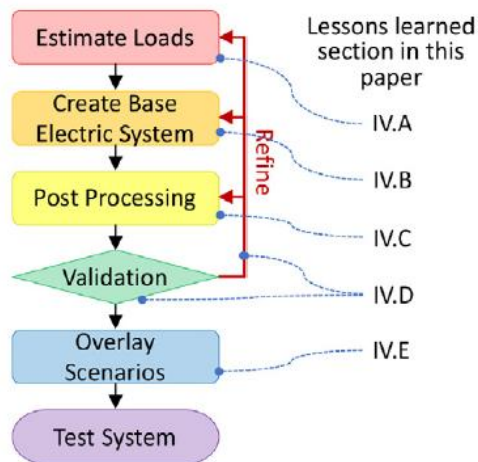


Fig. 4. The synthetic greater San Francisco Bay Area (SFO) distribution system [16].

Palmitier, Bryan, et al. "Experiences developing large-scale synthetic US-style distribution test systems." *Electric Power Systems Research* 190 (2021): 106665.

Geographic Information System (GIS)

What is GIS?

- A tool that connects data with geographic locations to show where and how things happen.
- Combines spatial data (maps, roads, terrain) with technical data (power grid, EV chargers, traffic).
- Help to visualize, analyze, and make location-based decisions.

Why GIS Matters for the ZEV Project?

- Shows where charging infrastructure, evacuation routes, and demand zones are located.
- Help to plan optimal deployment of mobile chargers and energy resources.
- Visualizes wildfire risk areas and their impact on grid resilience and evacuation safety.
- Integrates transportation, power, and hazard data into a single decision-making platform.

Geographic Information System (GIS)



ArcGIS

How We Use GIS in the ZEV Project?

- Overlaying power system nodes and load zones on real geographic maps.
- Mapping existing and planned EV charging stations for better infrastructure planning.
- Identifying critical routes during EV evacuation scenarios.
- Locating of mobile chargers and battery energy storage systems.
- Combining wildfire data, evacuation zones, and grid topology to enhance resilience.

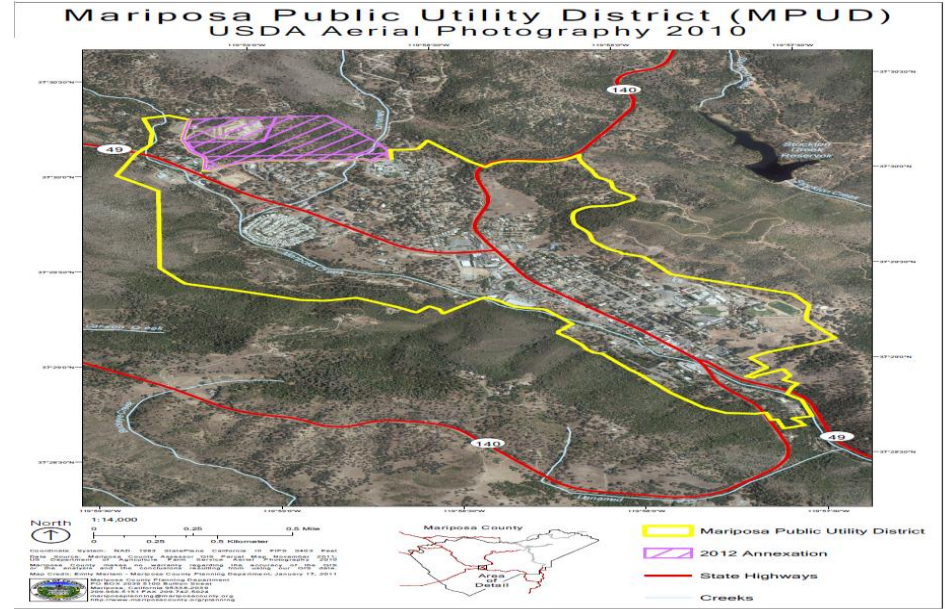
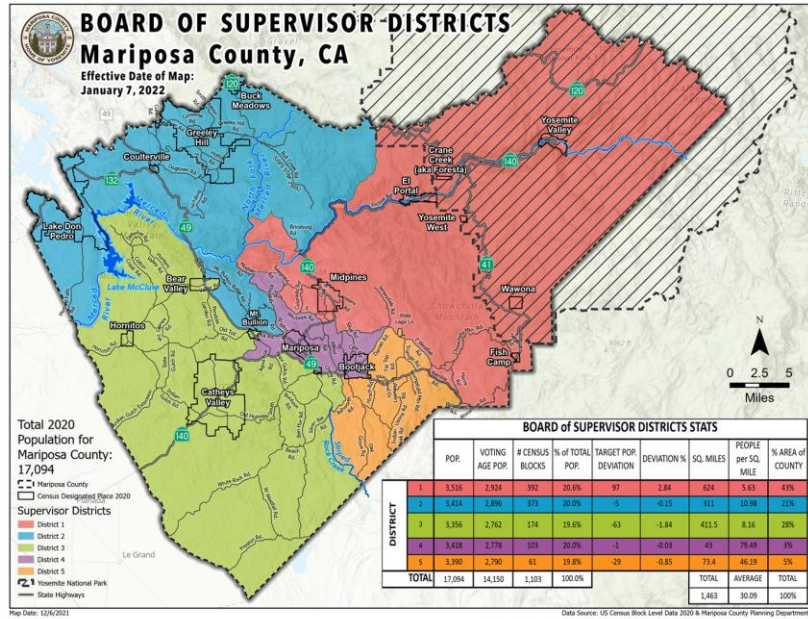


[Government](#)[Our Community](#)[How Do I](#)[Visitors](#)[Doing Business](#)[Maps](#)[Mariposa County Road Atlas](#)[For Use With Avenza PDF Maps](#)[Public Zoning Lookup Tool](#)[Home](#) » [Government](#) » [Planning](#) » [GIS \(Mapping\)](#) » [Maps](#)

Maps

- [Supervisory District Boundaries as of January 7, 2022](#)
- [25 Voting Precinct Maps](#)
- [Supervisory District Boundaries as of January 6, 2012 \(PDF\)](#)
- [Mariposa Town Plan: Design Review Overlay \(DRO\) and Historic Design Review Overlay \(HDRO\) Zoning Boundaries \(PDF\)](#)
- [Coulterville Town Plan: Design Review Overlay \(DRO\) and Historic Design Review Overlay \(HDRO\) Zoning Boundaries \(PDF\)](#)
- [Mariposa County Areas of Historic Fires and Prescribed Burns \(PDF\)](#)
- [Lake Don Pedro Community Service District \(PDF\)](#)
- [Yosemite Alpine Community Service District \(PDF\)](#)
- [Stockton Creek Preserve Trail Map \(PDF\)](#)
- [Upper Merced Watershed \(PDF\)](#)
- [Land Conservation Act Map A \(PDF\)](#)
- [Housing Project Fish Camp \(PDF\)](#)
- [IAP Hunter Incident \(PDF\)](#)
- [Historic Fires Prescribed Burns \(PDF\)](#)
- [Lushmeadows High Hazard Community Zone \(PDF\)](#)
- [Opportunity Zones for Mariposa County](#)

Case Study: Mariposa County



[chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.mariposacounty.org/DocumentCenter/View/90610/Supervisor-District-Boundaries-as-of-January-7-2022](https://efaidnbmnnnibpcajpcglclefindmkaj/https://www.mariposacounty.org/DocumentCenter/View/90610/Supervisor-District-Boundaries-as-of-January-7-2022)

Mariposa County: Land Use

i15 LandUse Mariposa1998

gis_admin
Private Organization

Summary

This data was developed to aid in DWR's efforts to continually monitor land use for the main purpose of determining the amount of and changes in water use.

[View Full Details](#)

Download

Details

 Dataset
Feature Layer

January 4, 2023
Info Updated

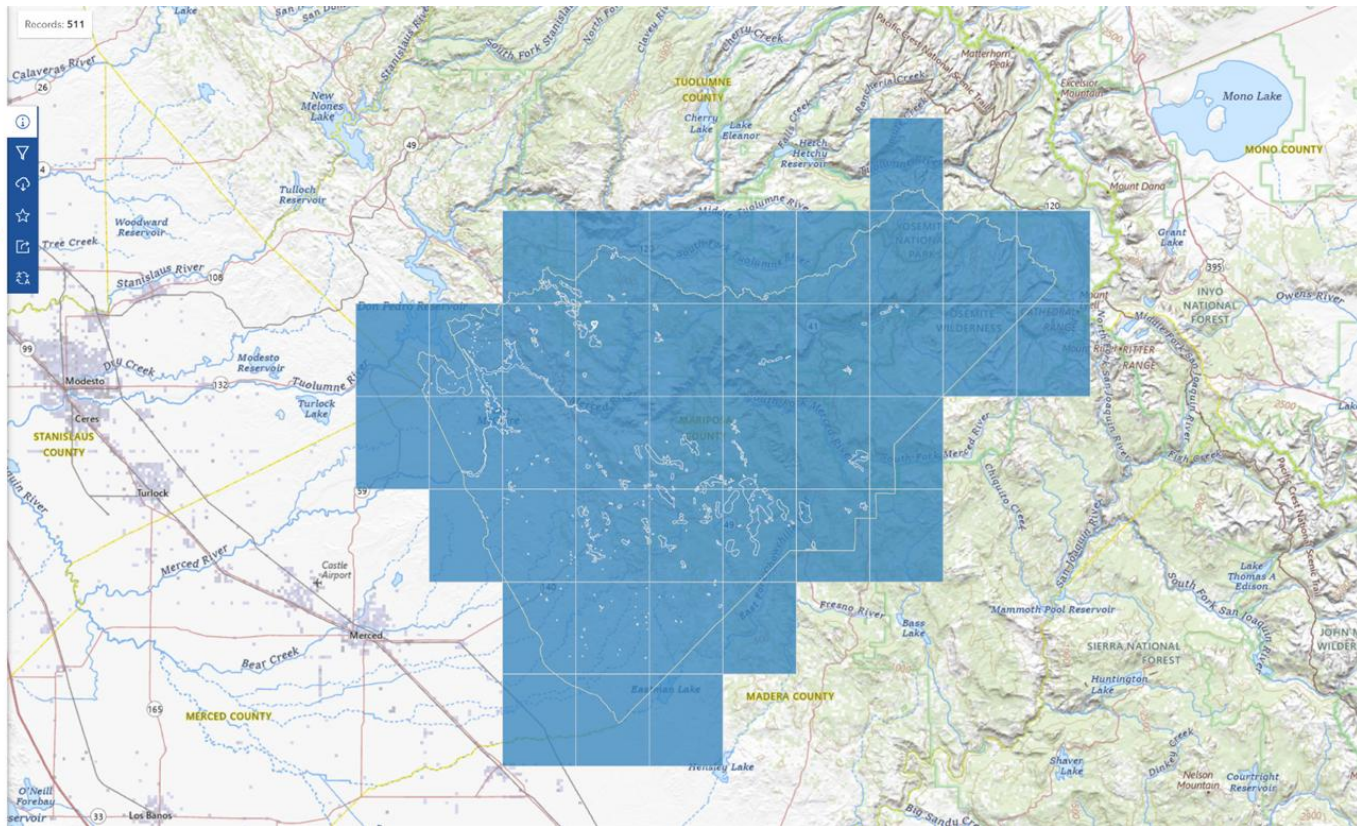
January 4, 2023
Data Updated

 January 4, 2023
Published Date

 Records: 511
[View data table](#)

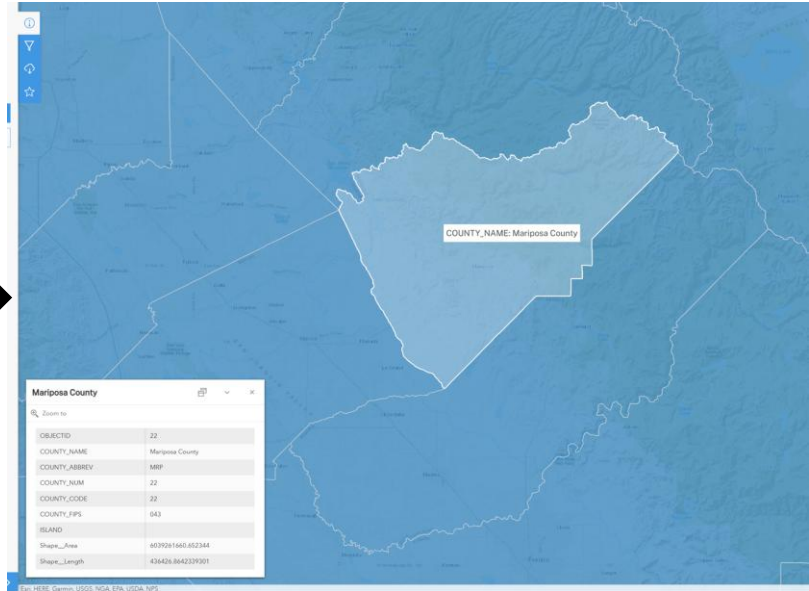
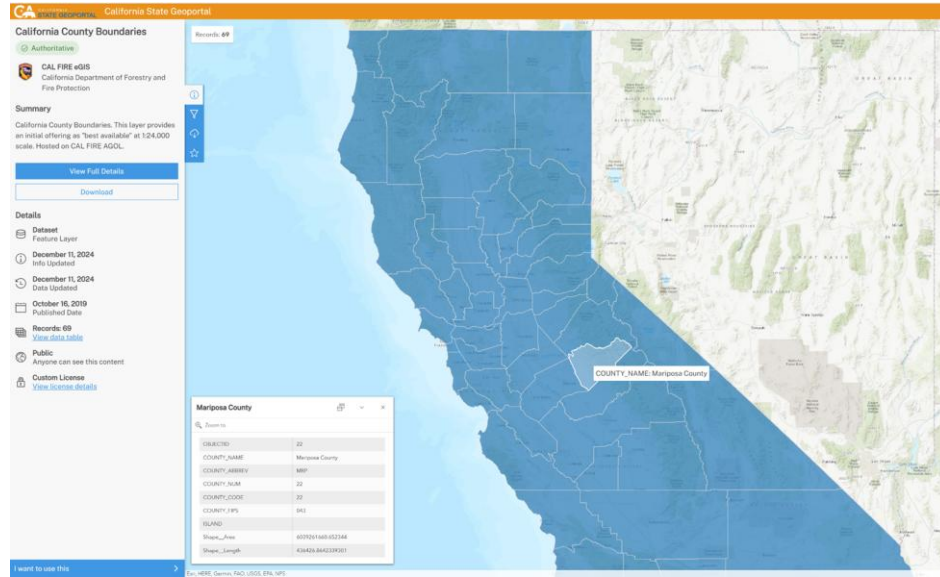
 **Public**
Anyone can see this content

 Custom License
[View license details](#)



Mariposa County: Boundary

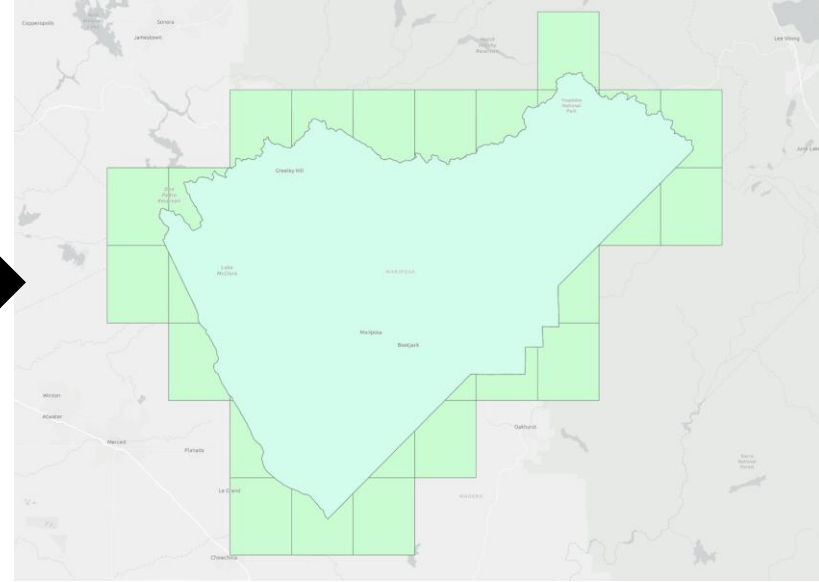
<https://data.ca.gov/dataset>



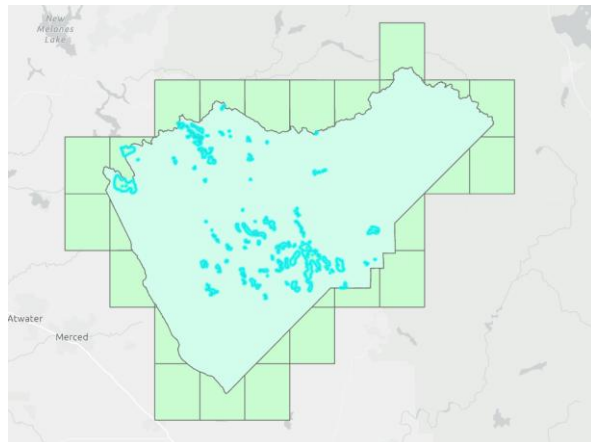
<https://gis.data.ca.gov/datasets/CALFIRE-Forestry::california-county-boundaries/explore?location=40.124447%2C-120.008073%2C8.26>

Mariposa: Land Use and Boundaries

ArcGIS Pro



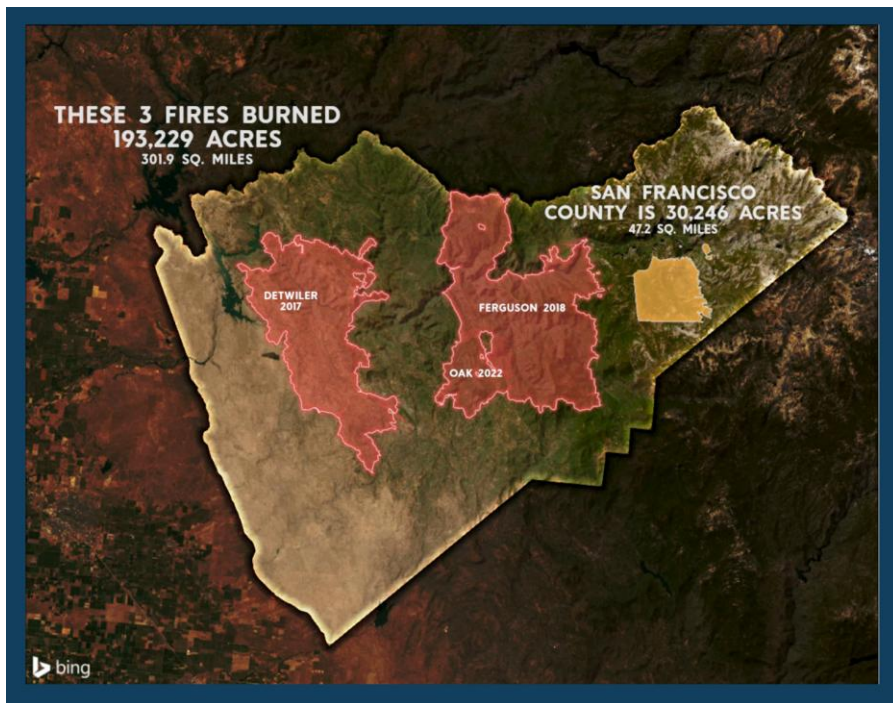
Load Zones: Specifications



BL_Y	ACRES	WATERSOURC	MULTUSE	CLASS1	SUBCLASS1	SPECOND1	IRR_TYP1PA	IRR_TYP1PB	PCNT1	CLASS2	SUBCLASS2	SPECOND2	IRR_TYP2PA	IRR_TYP2PB	PCNT2	CLASS3	SUBCLASS3	SPECOND3	IRR_TYP3PA	IRR_TYP3PB	PCNT3	UCF_ATT	SURVEYAREA	SURVEYYEAR	LABEL	Shape
1	419306	3385.44	4	S	Z	**	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SZ ***r00*****	MARIPOSA COUNTY	1998	nZ	22103
2	4193049	3755.758	4	S	NV	**	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SNW***r00*****	MARIPOSA COUNTY	1998	NV	2446
3	4185364	30276.169	4	S	Z	**	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SZ ***r00*****	MARIPOSA COUNTY	1998	nZ	19662
4	4179352	6373.036	4	S	NV	**	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SNW***r00*****	MARIPOSA COUNTY	1998	NV	4135
5	4181355	6.524	4	S	NW	**	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SNW***r00*****	MARIPOSA COUNTY	1998	NW	4
6	4180572	169.22	4	M	UR	**	*	n	*	10	NV	**	*	n	*	90	**	**	*	*	*	MUR***r10N6***r90...	MARIPOSA COUNTY	1998	UR+NV	108
7	4180572	73.045	*	S	P	4	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	S P 4***r00*****	MARIPOSA COUNTY	1998	nP4	47
8	4181032	29.159	4	S	NW	**	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SNW***r00*****	MARIPOSA COUNTY	1998	NW	16
9	4180284	237.126	*	S	P	4	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	S P 4***r00*****	MARIPOSA COUNTY	1998	nP4	152
10	4180917	20.164	*	S	P	4	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	S P 4***r00*****	MARIPOSA COUNTY	1998	nP4	13
11	4181009	4.053	4	S	S	1	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SS 1***r00*****	MARIPOSA COUNTY	1998	S1	2
12	4180457	52.387	*	S	P	4	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	S P 4***r00*****	MARIPOSA COUNTY	1998	nP4	34
13	4179421	29.752	*	S	P	4	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	S P 4***r00*****	MARIPOSA COUNTY	1998	nP4	15
14	4178684	294.158	*	S	P	4	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	S P 4***r00*****	MARIPOSA COUNTY	1998	nP4	150
15	4179432	1.878	4	S	S	1	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SS 1***r00*****	MARIPOSA COUNTY	1998	S1	1
16	4179352	10.477	*	S	D	1	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	S D 1***r00*****	MARIPOSA COUNTY	1998	nD1	6
17	4179121	10.181	4	S	S	1	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SS 1***r00*****	MARIPOSA COUNTY	1998	S1	
18	4178177	84.314	4	M	UR	**	*	n	*	10	NV	**	*	n	*	90	**	**	*	*	*	MUR***r10N6***r90...	MARIPOSA COUNTY	1998	UR+NV	54
19	4178016	36.77	*	S	P	4	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	S P 4***r00*****	MARIPOSA COUNTY	1998	nP4	2
20	4188084	19576.456	4	S	Z	**	*	n	*	00	**	**	*	*	*	**	**	*	*	*	**	SZ ***r00*****	MARIPOSA COUNTY	1998	nZ	12741

- Residential Load Zone
- Commercial Load Zone
- Critical Infrastructure
- Rural
- Transportation / EV Charging Zone

Wildfires in Mariposa

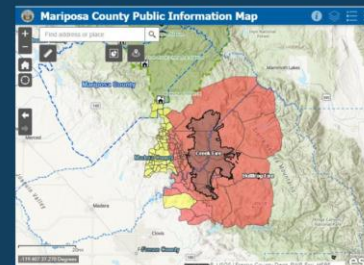


while Varney was engaged in constructing containment lines within a challenging and precipitous canyon. The Ferguson Fire consumed approximately 96,831 acres (151 sq. miles).

5. **River Fire (2021):** On August 10, 2021, the River Fire struck Mariposa & Madera County, causing further destruction and posing additional challenges for firefighters. The fire led to more evacuations and raised concerns about air quality in the region due to the dense smoke and falling ash. The River Fire destroyed 12 structures and burned approximately 9,655 acres (15 sq. miles).
6. **Oak Fire (2022):** The Oak Fire was a devastating wildfire that struck Mariposa County causing significant destruction and posing a serious threat to both residents and the environment. The fire erupted on July 22, 2022 and quickly spread across vast areas, fueled by dry vegetation, strong winds, and hot weather conditions. The fire's rapid expansion led to evacuation orders being issued for numerous communities in the affected region, forcing residents to flee their homes to safety. Firefighters and emergency responders worked tirelessly to contain the flames and protect property and lives. However, the wildfire's intensity and unpredictability made the firefighting efforts extremely challenging. The cause of the fire has been determined to be intentional arson and a local resident was arrested and charged with 4 felony counts on July 16, 2023. The Oak Fire destroyed 127 single resident structures and burned approximately 18,957 acres (30 sq. miles).
7. **French Fire (2024):** The French Fire started on July 4th, 2024 around 6:30 PM along French Camp Road near the Mariposa town edge. The fire burned 908 acres and destroyed 11 structures including some residential homes.

Honorable Mentions:

Rim Fire (2013)
Creek Fire (2020)



<https://storymaps.arcgis.com/stories/06d8fc4ad28d4e76bf7002fe35958b22>

Oak Fire: Report

4/27/25, 4:33 PM

Oak Fire: Incident Update on 08/11/2022 at 7:44 AM | CAL FIRE

Incident Update
Oak Fire

Date: 08/11/2022 Time: 7:44 AM

CAL FIRE MMU

Sierra NF

@CALFIREMMU

@SierraNF

@MariposaCountySheriff

Information Line: 1 (844) 668-3473

Media Line: (209) 628-3092

Incident Facts

Name Oak Fire	Start Date/Time 2022-07-22	Incident Status Active
Location Hwy 140 and Carstens Road, near Midpines	Type Wildfire	Cause Under Investigation
Counties Alameda, Mariposa	Administration Unit Unified Command: CAL FIRE Madera-Mariposa-Merced Unit and Sierra National Forest	
Unified Command Agency(s)	Size 19,244	Containment 98%
Structures Threatened 0	Structures Destroyed 193	Structures Damaged 10
Civilian Injuries	Firefighter Injuries 3	Civilian Fatalities
	Firefighter Fatalities	

Current Situation

Situation Summary

Minimal fire activity was observed today. Firefighters continued to mop up heavy, down fuels and hot spots. Hazard trees are being fell while suppression repair is ongoing.

Mariposa County Recovers Website:

For recovery assistance on for birth, death, or marriage certificates, insurance questions and disputes, replacement ID cards, vehicle registrations or titles, tax relief information, Veterans resources, home inspection experts, building, and more.

For more information visit: www.mariposacounty.org/265231/local-assistance-center

<https://www.fire.ca.gov/incidents/2022/7/22/oak-fire/updates/487160a8-3072-4a65-ad05-9dca2e01740>

100% Contained

19,244 Acres

1 County: Mariposa

Not a CAL FIRE Incident. Updates will be made as they become available.

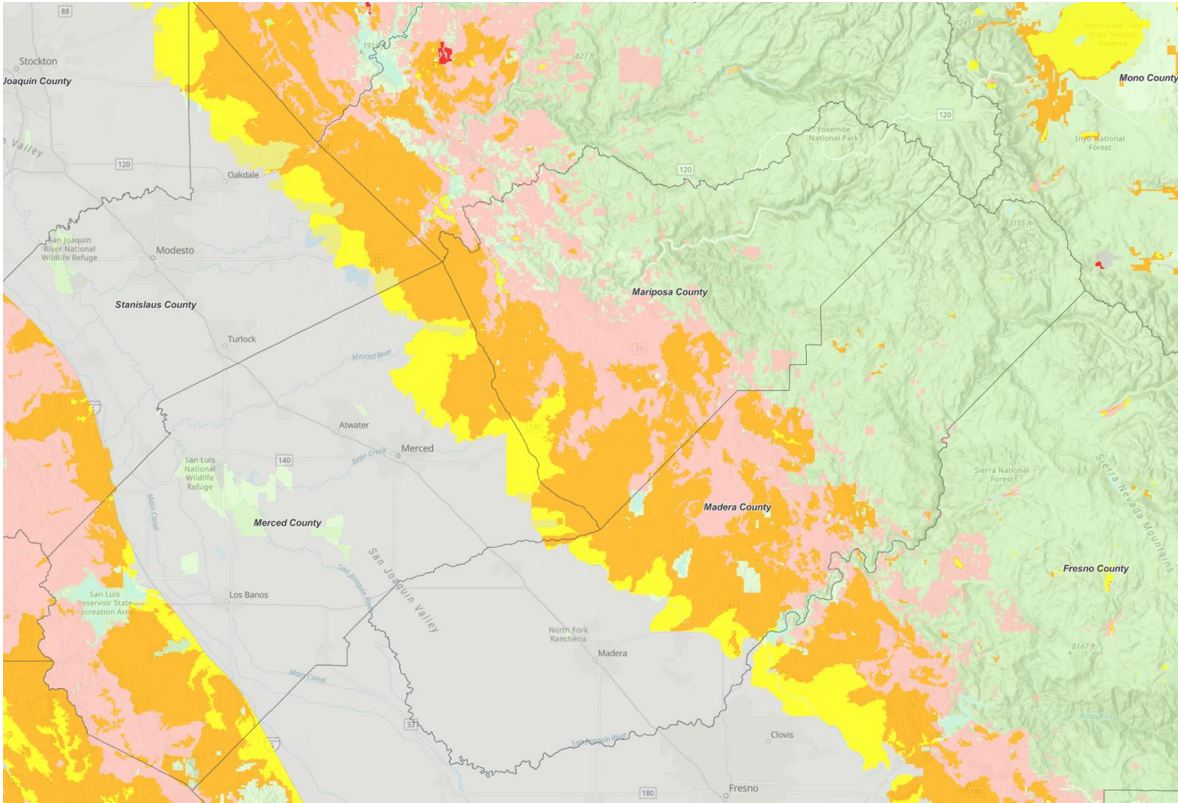
LAYERS

☒ Air Assets
 ☒ Evacuation Orders and Warnings
 ☒ Recent Perimeters
 ☐ Wind
 ☐ 5-Year Fire History

Date Started 07/22/2022 2:10 PM	Last Updated 06/16/2023 9:53 AM	Cause Arson
Incident Maps and Reports 3D Map	Admin Unit CAL FIRE Madera-Mariposa-Merced Unit and Sierra National Forest	Location Hwy 140 and Carstens Road, near Midpines [37.5509366,-119.9234728]
External Incident Link https://incivweb.nwcg.gov/incident/8280/	Reports Status reports	

<https://www.fire.ca.gov/incidents/2022/7/22/oak-fire>

Fire Hazard Severity Zone Viewer



About this Map

This map displays adopted Fire Hazard Severity Zones (FHSZ) in the State Responsibility Area (SRA), effective April 1st 2024. It also displays recommended FHSZ in the Local Responsibility Area (LRA) from 2007-2011.

Due to regulatory processes, there are lands that are no longer classified as SRA and have become classified as LRA yet had a FHSZ designation from the 2007 SRA FHSZ map adoption. These areas are shown on the map with the same symbology.

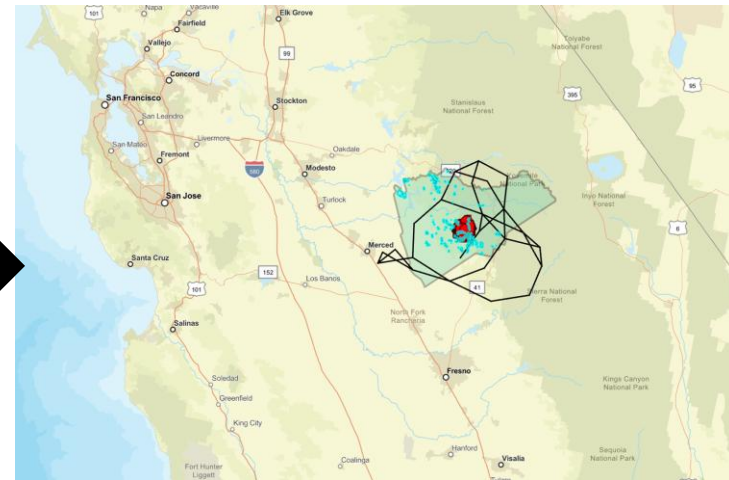
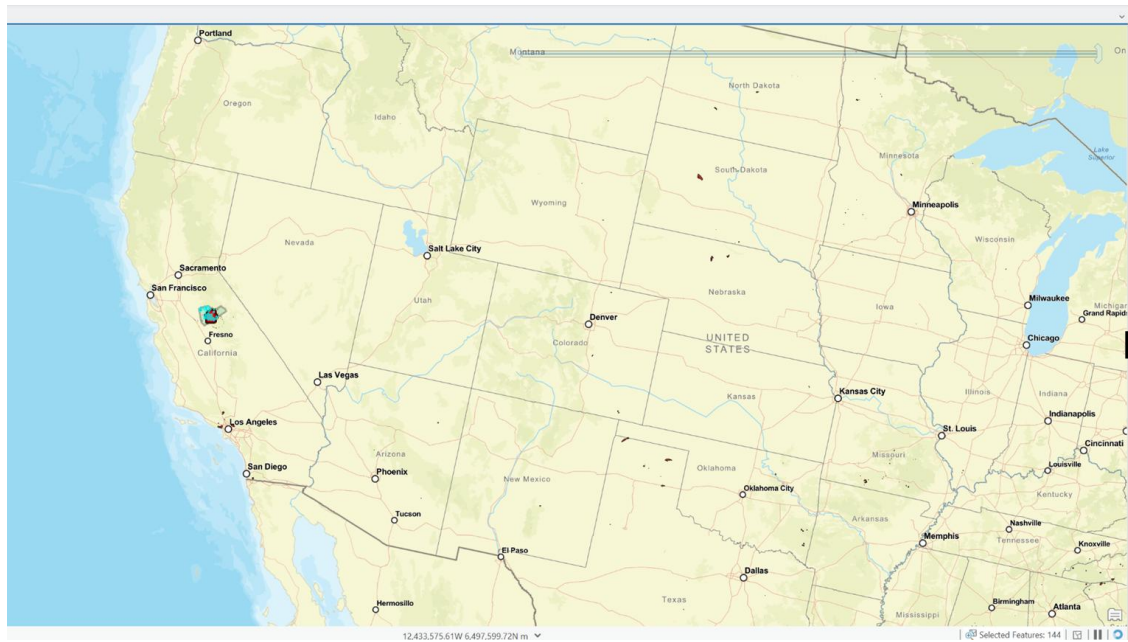
To verify your Fire Hazard Severity Zone in LRA, please contact your local agency.

Use the link below to access the combined FHSZ data:

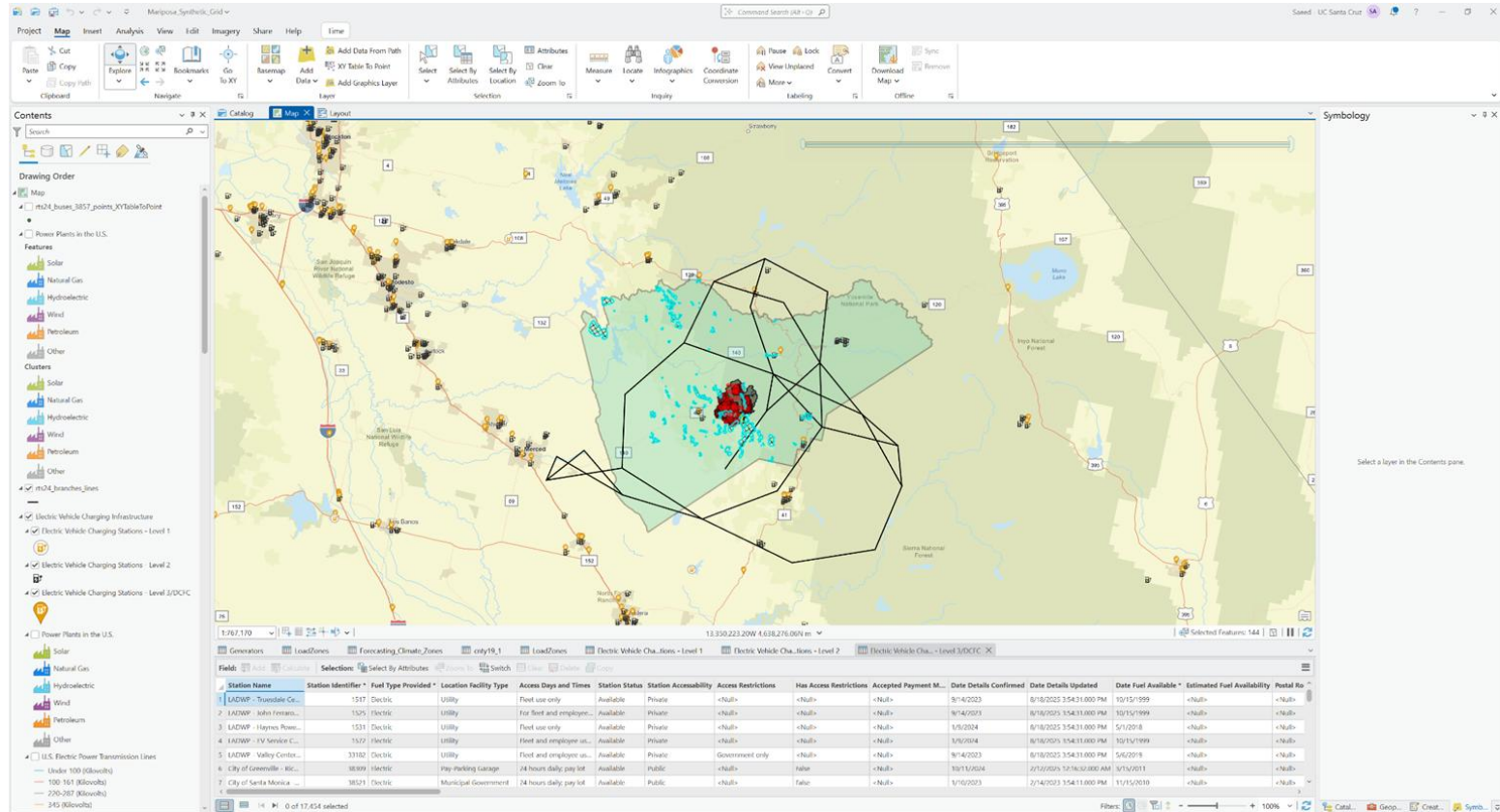
[LINK](#)

Legend	Map Layers
Fire Hazard Severity Zones	
FHSZ in SRA - Effective April 1, 2024	
Very High	
High	
Moderate	
FHSZ in LRA - Reclassified from SRA	
Very High	
High	
Moderate	
FHSZ in LRA - Recommended 2007-2011	
Very High	
State Responsibility Areas	
Federal Responsibility Area (FRA)	
Local Responsibility Area (LRA)	
State Responsibility Area (SRA)	

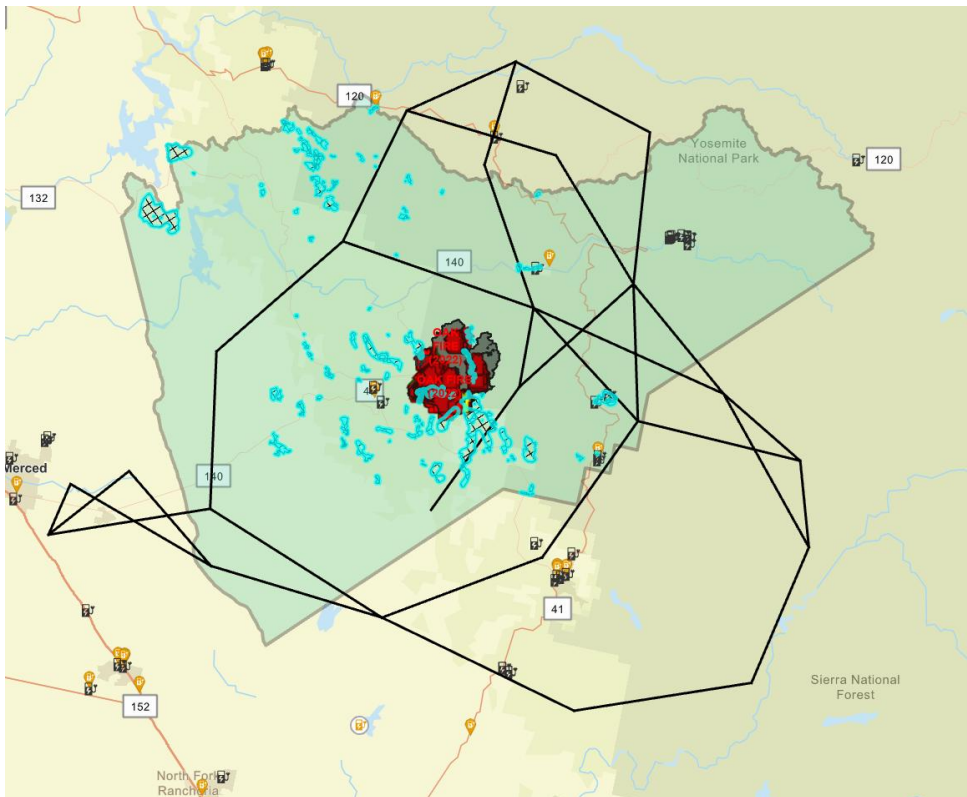
ArcGIS: Overview



ArcGIS: Mariposa (Overlay of the IEEE Power Grid)

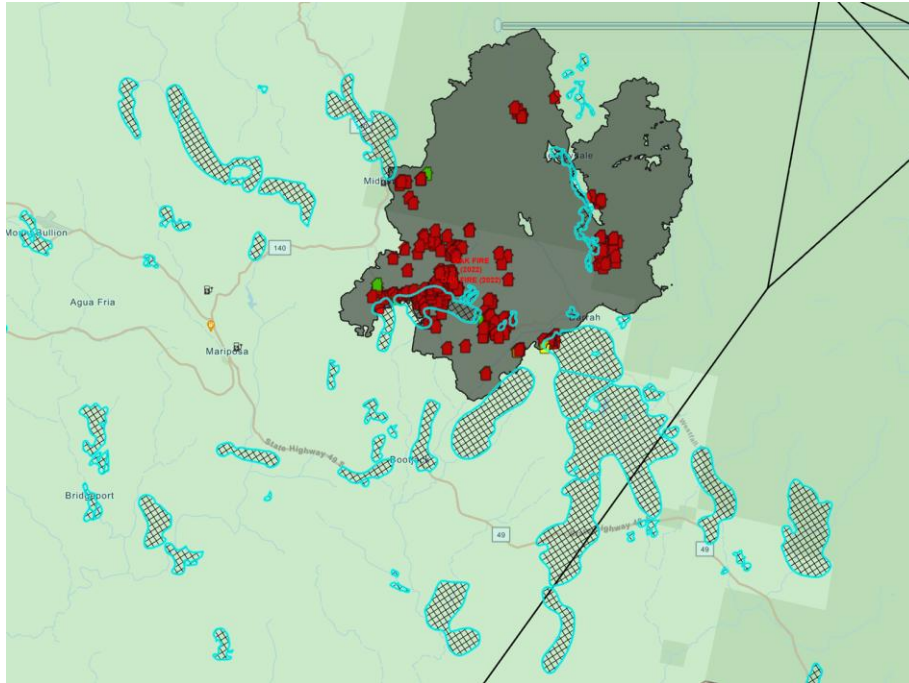


IEEE Power Grid Overlay: Charging Stations



- ✓ Electric Vehicle Charging Infrastructure
 - ✓ Electric Vehicle Charging Stations - Level 1
 - 
 - ✓ Electric Vehicle Charging Stations - Level 2
 - 
 - ✓ Electric Vehicle Charging Stations - Level 3/DCFC
 - 

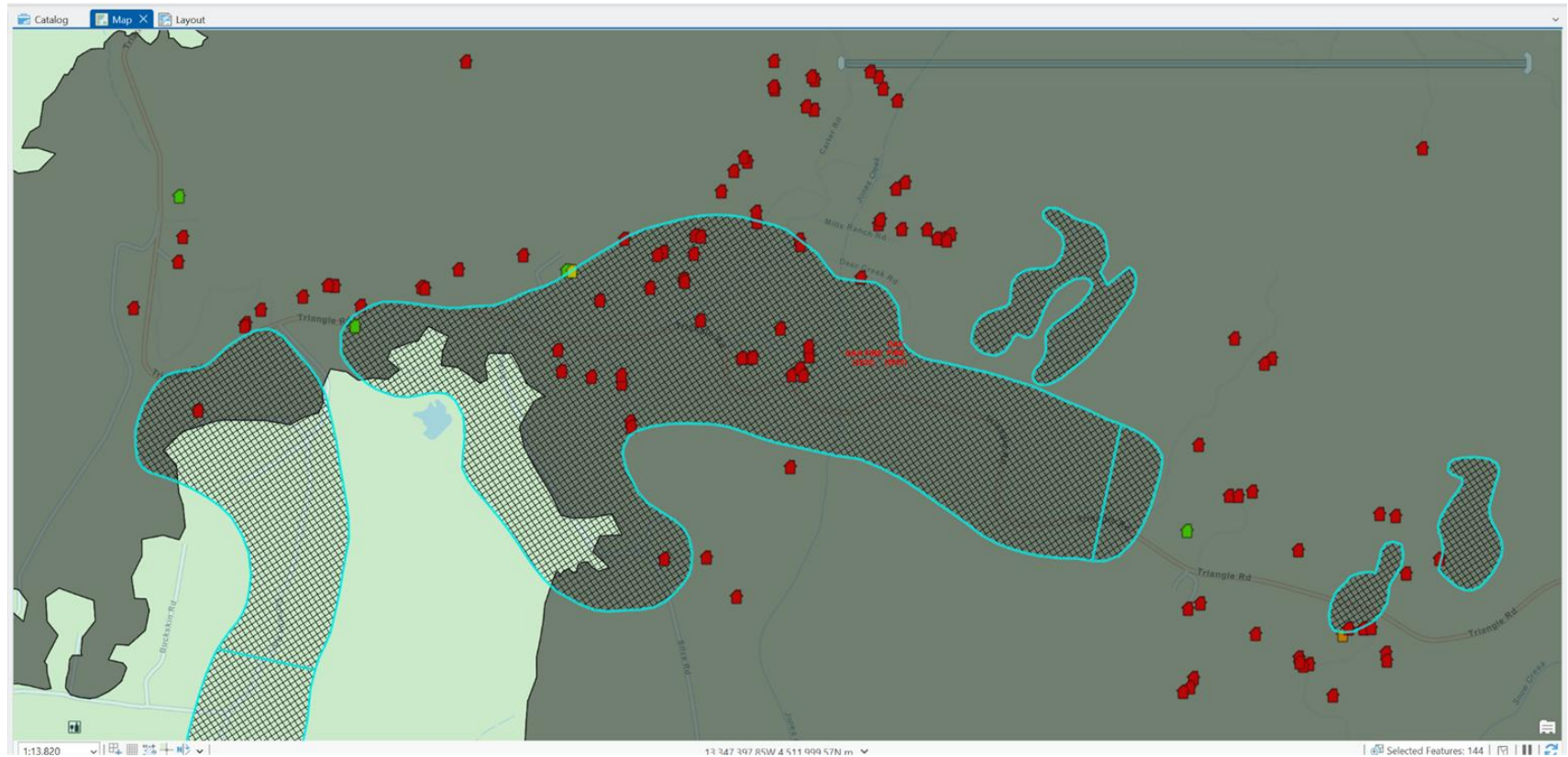
ArcGIS: Oak Fire (2022)



☑ Oak_2022_DINS_Public_View

- Destroyed (>50%)
- Major (26-50%)
- Minor (10-25%)
- Affected (1-9%)
- Inaccessible
- No Damage

ArcGIS: Load Zones and Oak Fire

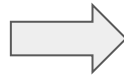


Load Estimation

PCNT3	UCF_ATT	SURVEYAREA	SURVEYYEAR	LABEL	Shape_Are	Shape_Len	atlas_wmas
**	MUR***n*10NV***n*90*...	MARIPOSA COUNTY	1998	UR+NV	1099990.63433	8420.341101	685528.651855
**	S P 4*n*00*****...	MARIPOSA COUNTY	1998	nP4	473724.944412	4317.511445	295237.76816
**	S P 4*n*00*****...	MARIPOSA COUNTY	1998	nP4	1529769.22136	9568.661769	953479.701957
**	S P 4*n*00*****...	MARIPOSA COUNTY	1998	nP4	130116.642636	2234.216498	81085.605032
**	S P 4*n*00*****...	MARIPOSA COUNTY	1998	nP4	340579.297247	2930.386912	212269.069741
**	S P 4*n*00*****...	MARIPOSA COUNTY	1998	nP4	195227.997412	2821.956874	121703.945516
**	S P 4*n*00*****...	MARIPOSA COUNTY	1998	nP4	1906338.22729	11931.346952	1188648.0275
**	MUR***n*10NV***n*90*...	MARIPOSA COUNTY	1998	UR+NV	549306.649149	3079.081026	342541.109422
**	S P 4*n*00*****...	MARIPOSA COUNTY	1998	nP4	231726.91132	2350.596259	144507.061422
**	MUR***n*10NV***n*90*...	MARIPOSA COUNTY	1998	UR+NV	297406.239391	2597.830123	185139.665664

$$\text{Area}_{\text{m}^2} = 1,099,990$$

$$R = 10,000 \text{ kWh/year/acre}$$



$$\text{Estimated Load} = 271.8 \times 10,000 = 2,718,000 \text{ kWh/year}$$

$$\text{Area}_{\text{acre}} = \frac{1,099,990}{4046.86} \approx 271.8 \text{ acres}$$

Load Estimation

Sum of Estimated Load → Total Demand for Mariposa County (in kWh/year)

Convert to MW:

$$\text{Peak Demand (MW)} \approx \text{Total_Demand (kWh/year)} / (8760 \times \text{LF})$$

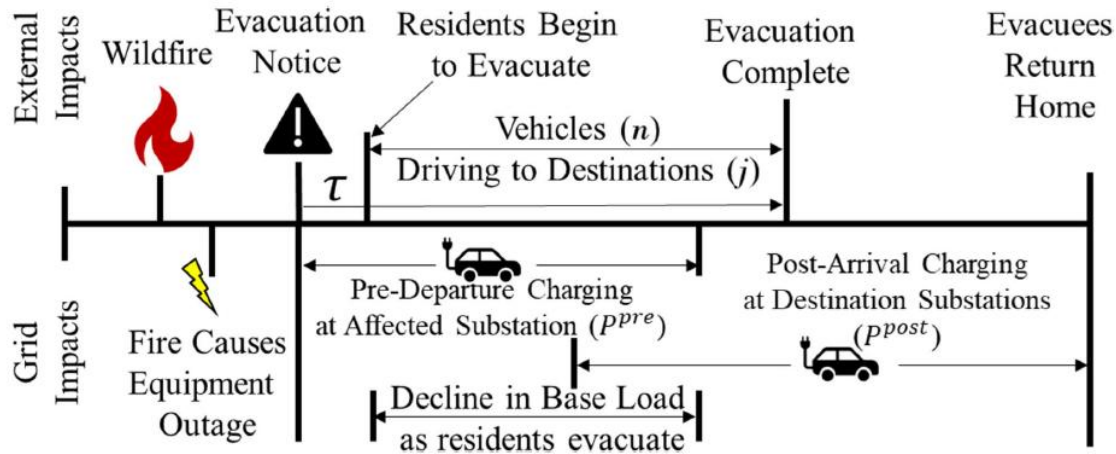
(LF = load factor, e.g., 0.5–0.6)

	A	B	C	D	E	F
1	LABEL	Shape__Are	Area_acres	Demand_Rate_kWh_per_acre	Estimated_Load_kWh_per_year	
2	UR+NV	1099990.63	271.8133639	10000	2718133.639	
3	UR+NV	1527692	377.5005807	10000	3775005.807	
4	nP4	800000	197.6841304	5000	988420.6521	
5	nP4	1300000	321.2367119	5000	1606183.56	
6	UR+NV	500000	123.5525815	10000	1235525.815	
7	nP4	700000	172.9736141	5000	864868.0706	
8	TOTAL	5927682.63	1464.760983		11188137.54	
9						
10						
11						
12						
13						

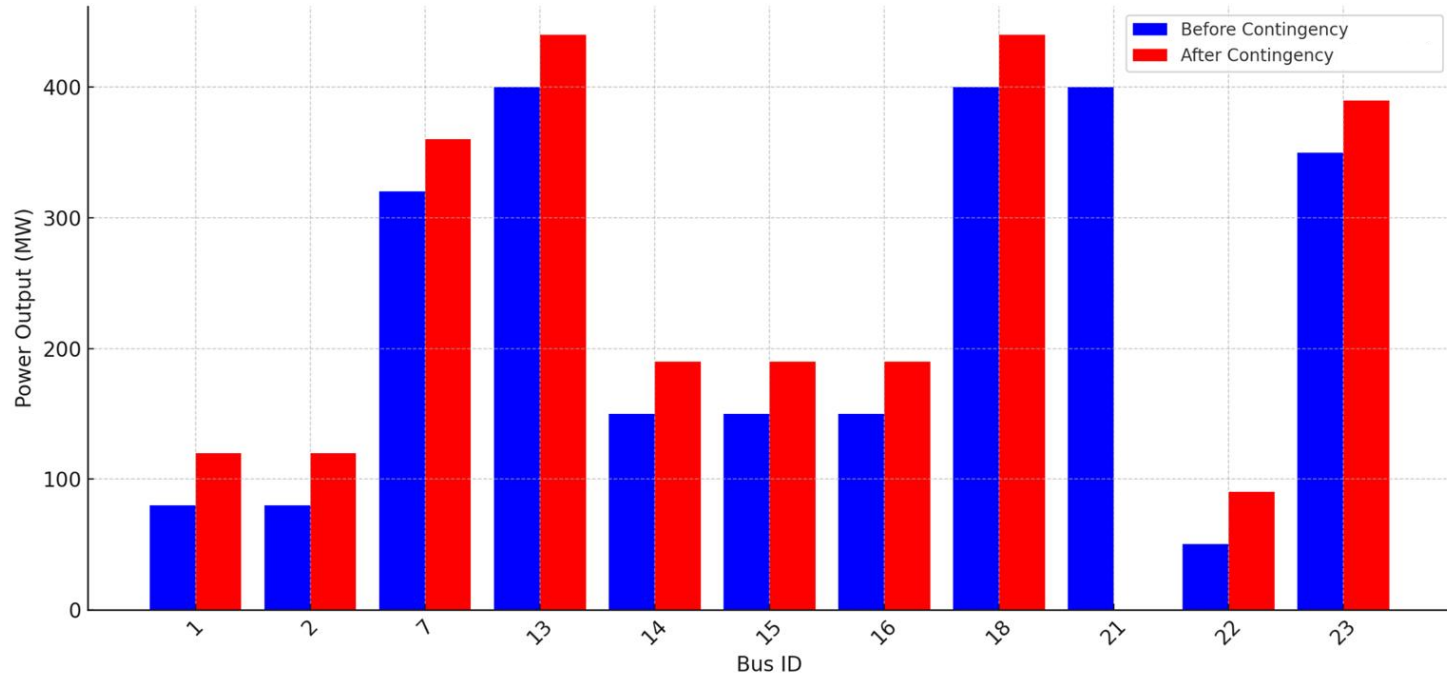
Evacuation Algorithm

Include **pre-departure** EV charging demand at local substations.

Model **post-arrival** charging demand at destination substations.

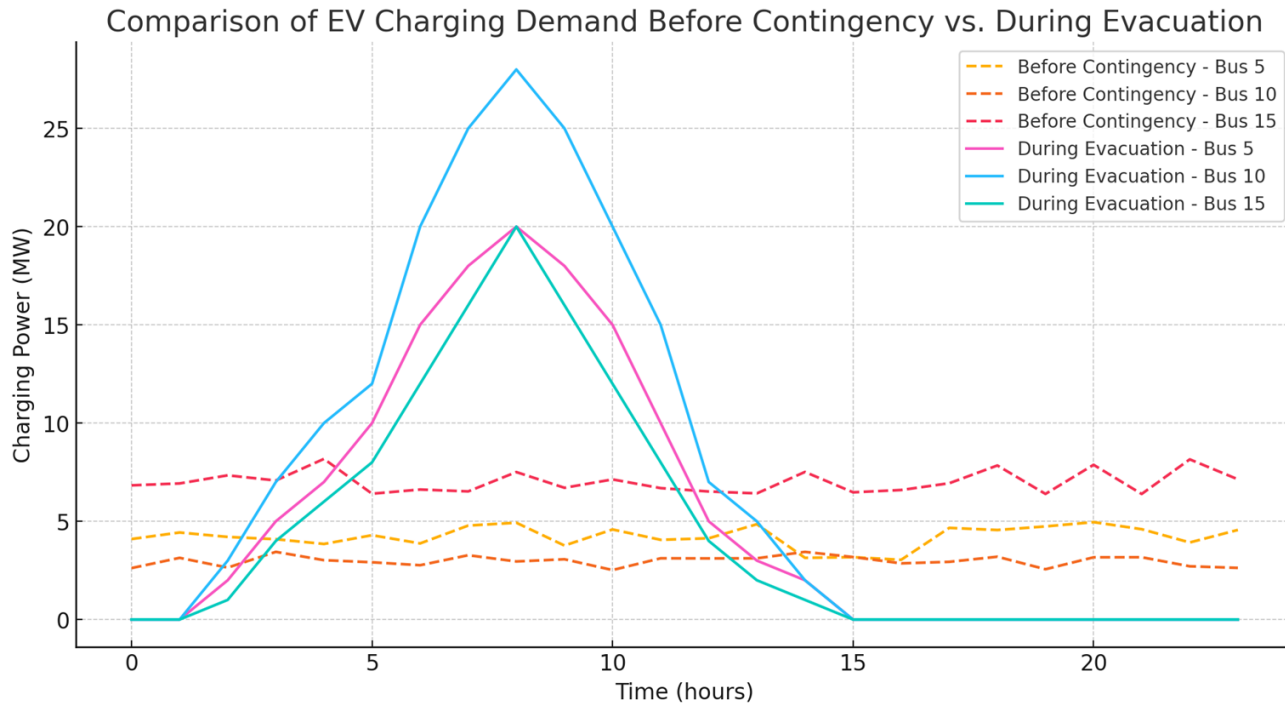


Optimal Power Flow: N-1 Criterion (Security)



Generator Failure: #21

Evacuation Scenario



Resource Allocation

Mobile Chargers and Mobile Battery Energy Storage System

Objective function 1: $\min \left(\sum_{v \in \mathcal{V}} \sum_{t \in \mathcal{T}} y_{vt}^{\text{EV}} + \sum_{n \in \mathcal{N}} \sum_{t \in \mathcal{T}} y_{nt}^{\text{grid}} \right)$

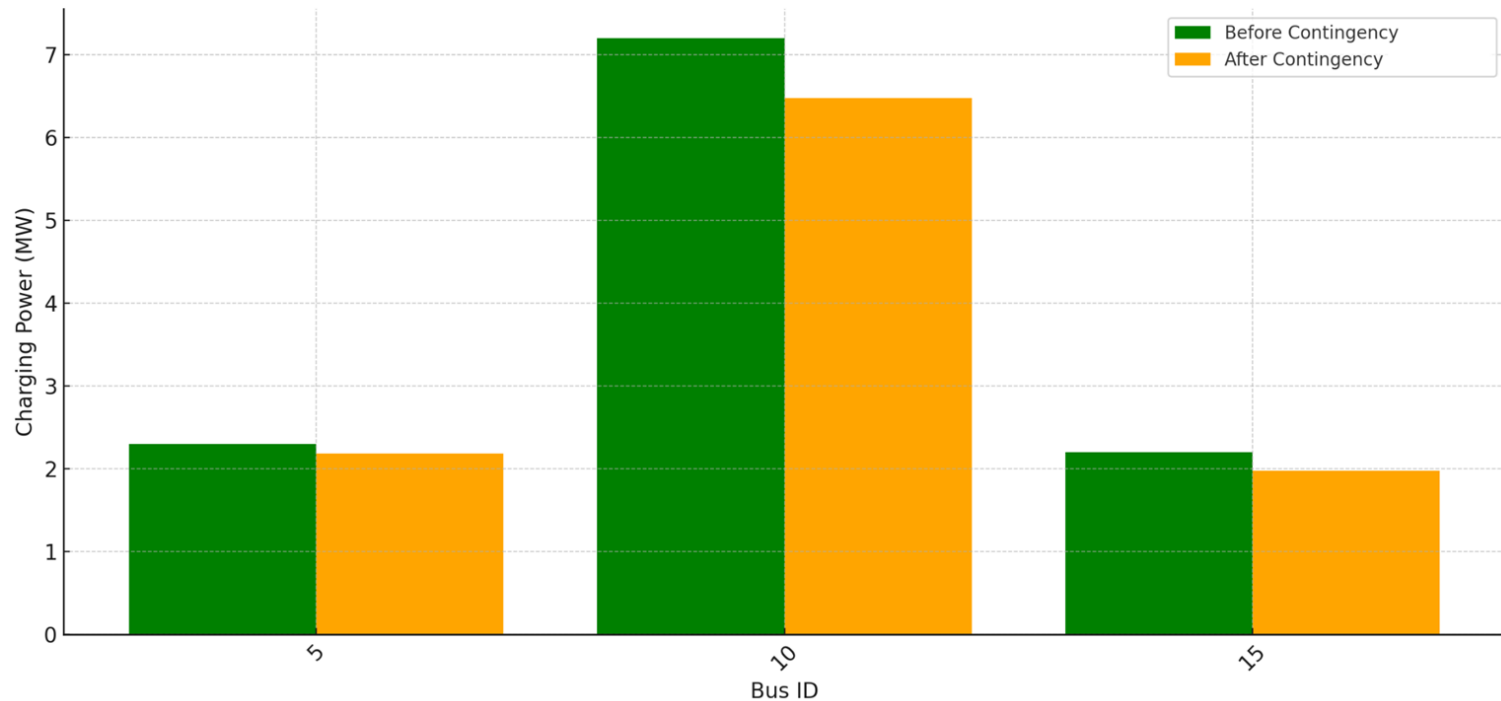
Objective function 2: $+ \alpha_1 \sum_{m,n,t} x_{mnt} \cdot d_{mn} + \alpha_2 \sum_{s,n,t} z_{snt} \cdot d_{sn}$

Objective function 3: $-\beta \sum_{s,n,t} p_{snt}^{\text{dis}} - \gamma \sum_{m,n,t} p_{mnt}^{\text{EV}}$

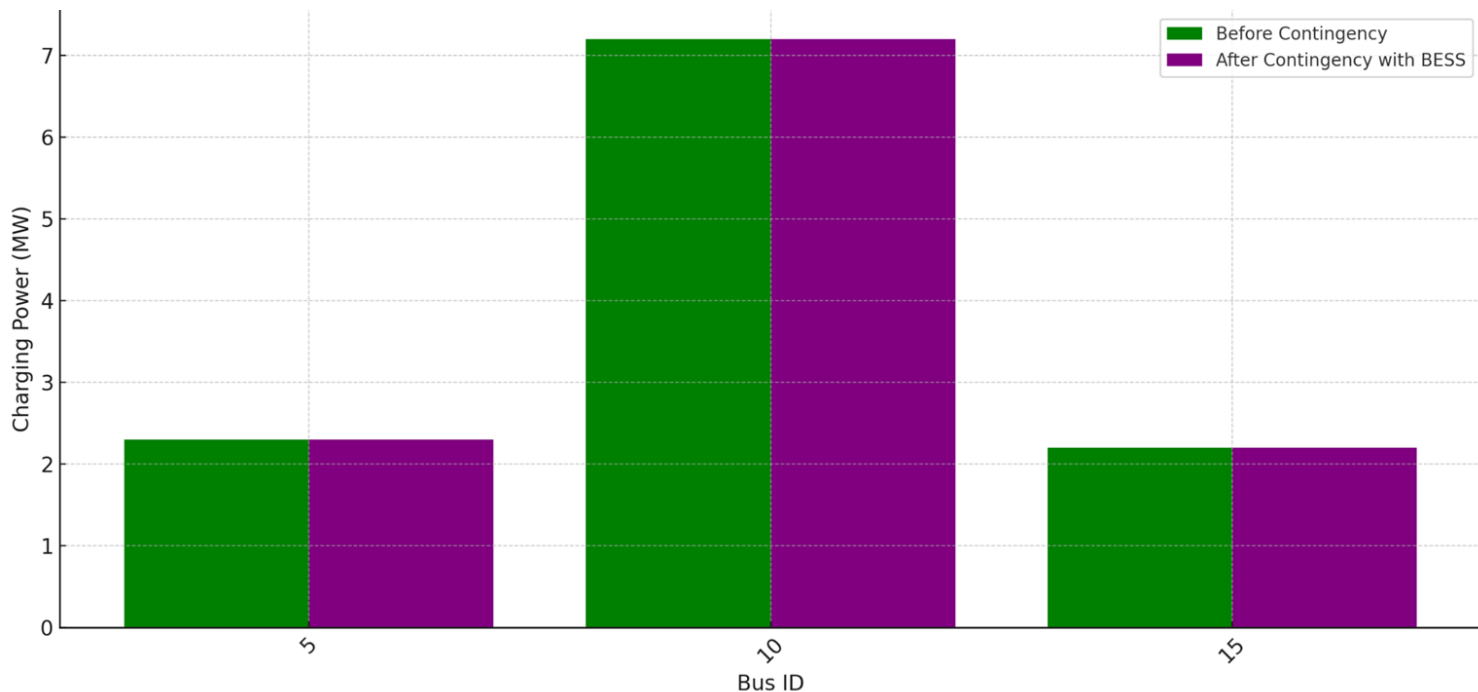
Objective Term	Purpose
$\sum y_{vt}^{\text{EV}}$	Minimize unserved EV charging demand
$\sum y_{nt}^{\text{grid}}$	Minimize unserved grid load
$\sum x_{mnt} \cdot d_{mn}$	Minimize mobile charger deployment cost
$\sum z_{snt} \cdot d_{sn}$	Minimize mobile ESS relocation cost
$\sum p_{snt}^{\text{dis}}$	Maximize energy delivered to grid
$\sum p_{mnt}^{\text{EV}}$	Maximize energy delivered to EVs

Objective: Minimize unserved energy, minimize deployment cost, maximize the resilience

Charging Stations



Mobile Chargers/Energy Storage



Transportation

Step 1: Evacuation demand

- Residents evacuate along highway like 140.
- Departure pattern: Time-dependent EV flows.

Step 2: Arrival at charging stations

- Evacuees: move and stop at nearest charging stations (blue nodes).

Step 3: Coupling to power system

- Each charging station is mapped to a bus in the synthetic grid.
- Converting transportation demand into time-varying bus load injection in the power system model.

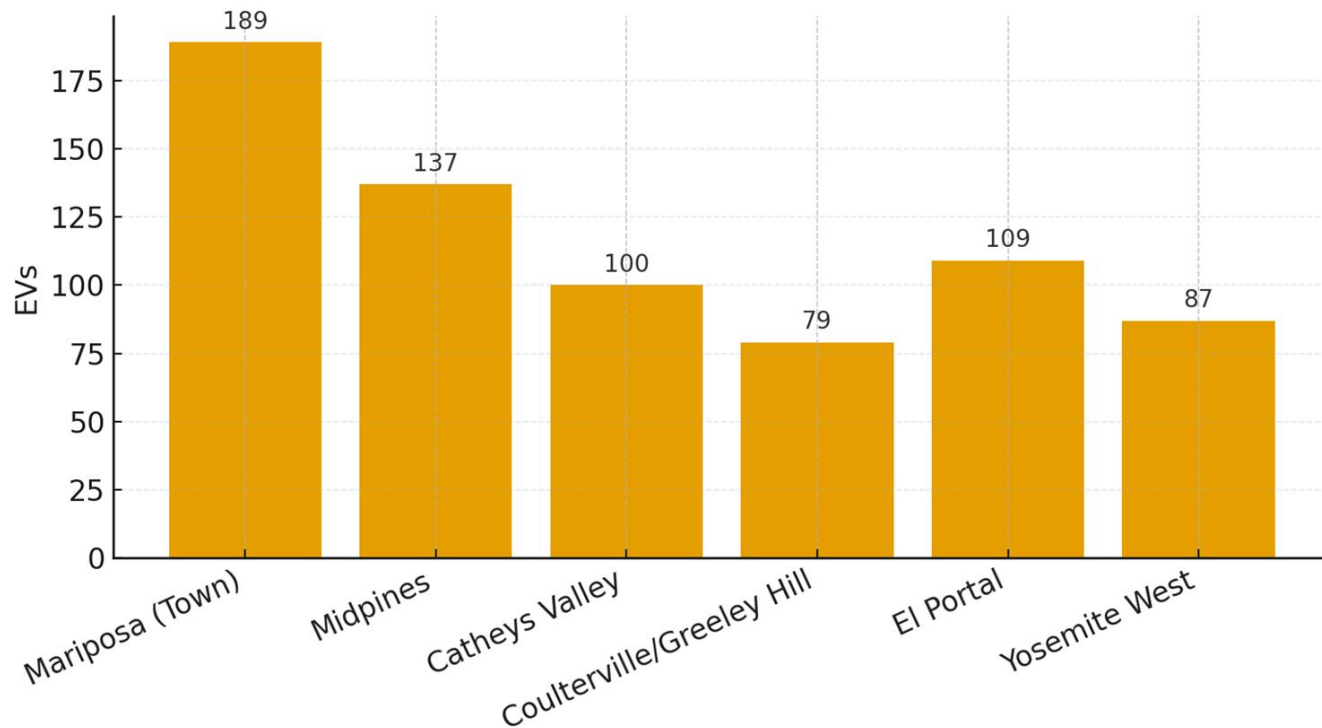
Approach: Origin–Destination

Origin (O): Where evacuees start their trip.

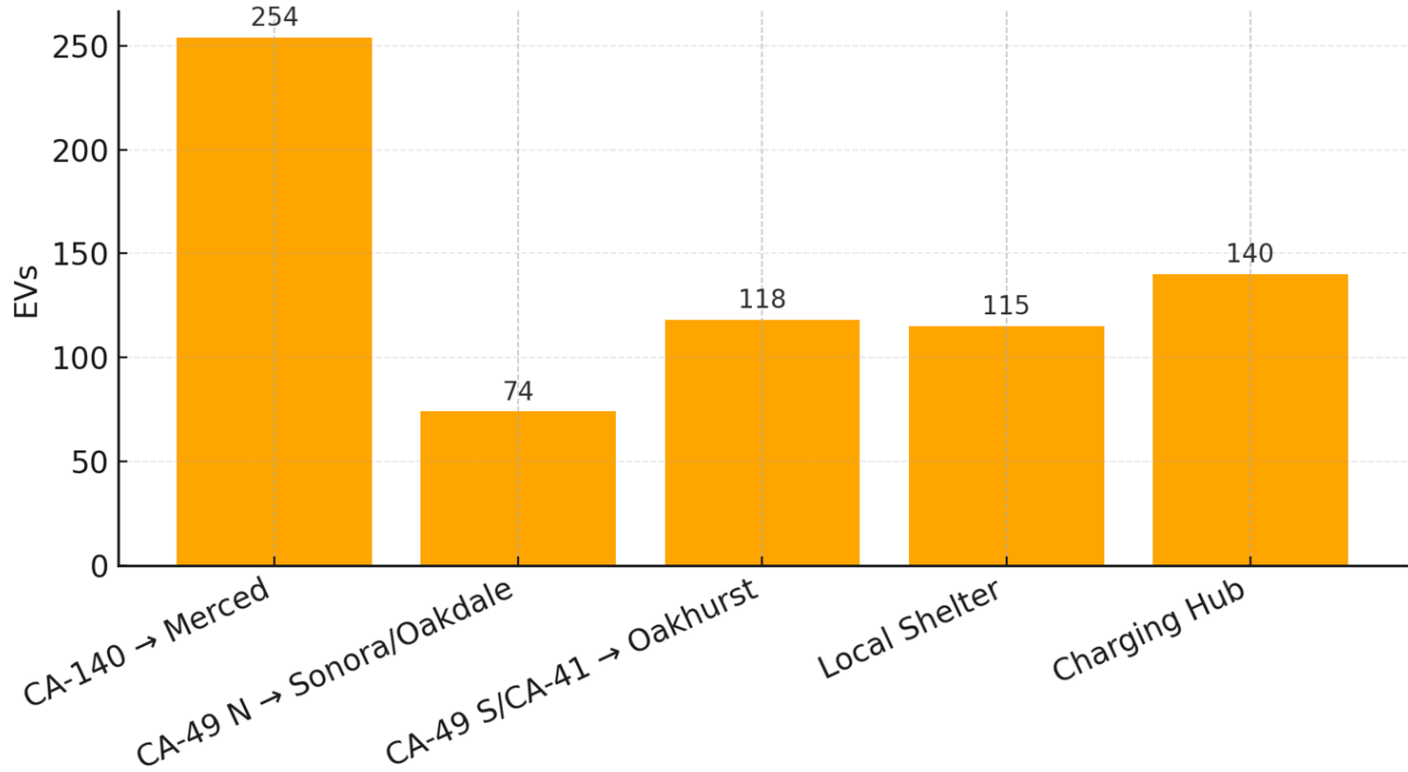
Destination (D): Where evacuees are trying to reach safely.

OD pair: A link between a starting point and an endpoint.

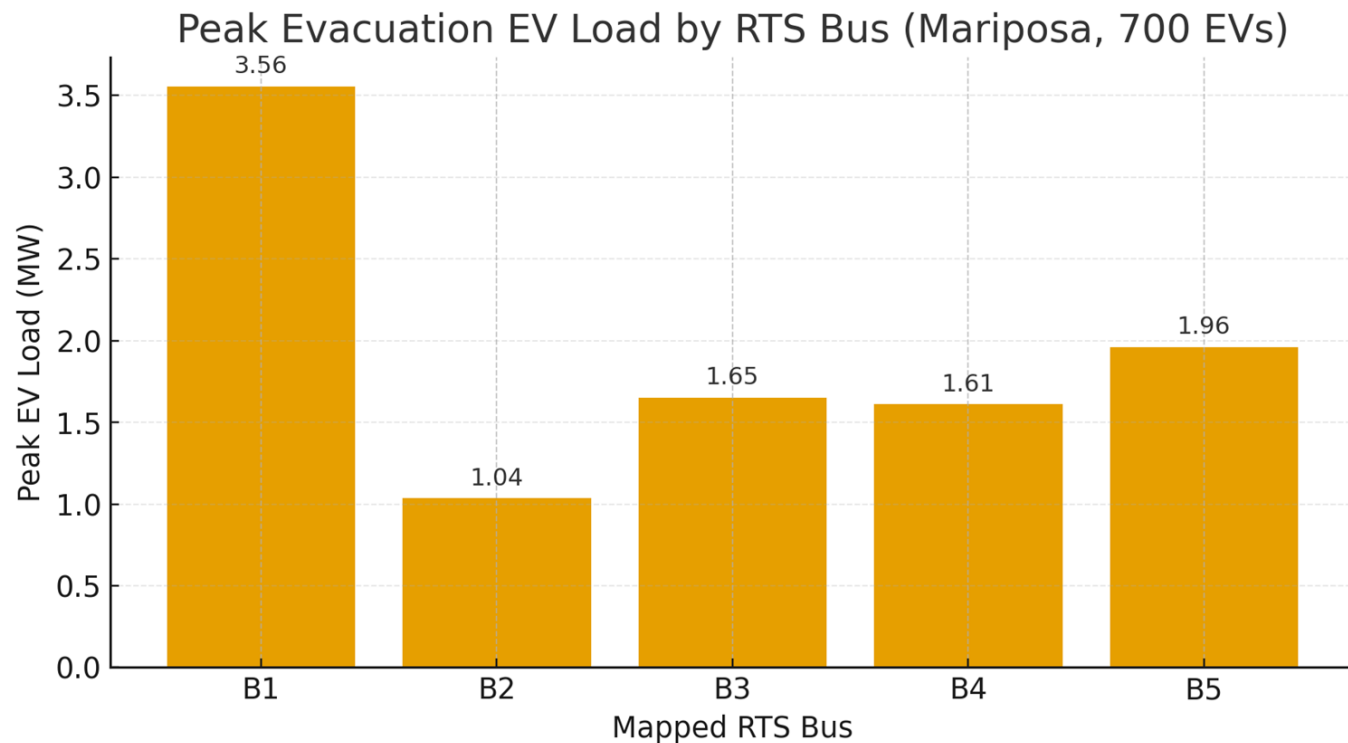
Origin



Destination



Translate to Power Systems



Conclusion

- Developed an integrated framework that couples **power system optimization, transportation modeling, and GIS-based analysis** to support ZEV evacuation planning.
- **Optimal Power Flow (OPF)** and **N-1 contingency** analysis can improve grid resilience during high-demand evacuation scenarios.
- Built a **synthetic power grid** for **Mariposa County** with spatially-linked load zones, charging stations, and wildfire data for decision-making.
- Showed that **mobile chargers and BESS** units enhance system reliability and reduce unserved energy during emergencies.

Future Directions

- Incorporate **uncertainty modeling** (stochastic demand, infrastructure failures) into the optimization.
- Extend the model to **bi-level** planning for long-term infrastructure investment and short-term operational decisions.
- Validate the approach with **real-world data** and potential deployment scenarios in collaboration with agencies and utilities.

Q&A?

Thanks for your attention!



 saliadoo@ucsc.edu